



THULANI MAGEDARA

AI DATA INTELEGENCE ASSOCIATE

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ABOUT ME

Innovative and adaptable developer skilled in Java, React, Spring Boot, and SQL/No SQL, with hands-on experience in building full-stack applications, IoT systems, and AR-based solutions. Passionate about crafting efficient, user-focused software while thriving in collaborative, fast-paced environments. Known for strong problem-solving, leadership, and the ability to turn ideas into impactful digital solutions.

EDUCATION

BSc (Hons) in Electronics and Computer Science	2026 Graduation
University of Kelaniya	
GPA: 3.55	
Relevant Coursework:	
• Software Engineering • Database Management System • Machine Learning & AI • IoT Systems & Embedded Systems • Industrial Automation	• Enterprise Application Development with Java • Data Structures and Algorithms • Object-Oriented Programming with Java • Industrial Electronics

EXPERIENCE

	Applications Engineer - Intern Protonest IOT Pvt. Ltd	6 months
Designed and developed high-fidelity IoT ecosystems to accelerate client acquisition, focusing on seamless frontend interfaces and robust firmware testing.		
• Frontend & UI/UX: Designed Userflow Diagrams and developed mobile-responsive dashboards for three flagship apps (Plant, Factory, and Fleet Management) using React.js, Tailwind CSS, and Figma. • Firmware & Testing: Developed ESP32 firmware for concurrency testing and used MQTTX to validate MQTT topics and protocols. • Tools & Monitoring: Conducted API testing via Postman, visualized telemetry in Grafana, and created technical demo videos for client pitches. • Deployment: Managed application deployment and maintained CI/CD pipelines using GitHub Actions to automate builds, testing, and delivery, ensuring rapid and reliable software releases. • AI-Enhanced Development: Leveraged GitHub Copilot and Claude (Sonnet/Opus/GPT) using Agent mode, Planning, and RAG architectures to optimize code efficiency and system architecture. • Tech Stack & IDEs: Worked across VS Code, Arduino IDE, WebSockets, and REST protocols.		

PROJECTS

Dashboard Development for IOT Systems	Protonest IOT Pvt Ltd
• Developed three mobile-responsive dashboard applications (Plant, Factory, and Fleet Management) featuring real-time data visualization via WebSocket connections, Historical trends, and actuator control via REST APIs. • Engineered ESP32 firmware for concurrency testing and utilized MQTTX to validate MQTT protocols, ensuring seamless live telemetry between hardware and frontend. • Implemented automated CI/CD pipelines using GitHub Actions for streamlined deployment and utilized AI Architectures (Agent Mode, Planning, RAG) to optimize system logic. • Tech Stack: React.js, Tailwind CSS, ESP32, MQTT, WebSockets, REST, GitHub Actions, Postman, Grafana, Claude (Sonnet/Opus/GPT), GitHub Copilot.	

“Sigiri Yathra” - Augmented Reality for Cultural Heritage: Reconstructing the Lost Kingdom of Sigiriya

UOK

- The app uses the camera to recognize key landmarks in Sigiriya, such as the Lion’s Paw Entrance, Audience Hall, Royal Palace Ruins, Water Gardens, Octagonal Pond, and Dagoba.
- Once recognized, it displays a 3D reconstruction of the site along with an information panel so users can explore how these places looked in ancient times.
- **Tech Stack:** Unity, C#, ONNX, Unity Barracuda, Blender, Agisoft Metashape

Fetch Me Home - Web-based Pet Adoption platform

UOK

- This connects users with animal shelters, enabling users to browse pets by type, breed, and location, view detailed profiles, and submit adoption requests, while shelters manage listings and requests through an admin dashboard.
- **Tech Stack:** React.js, Spring Boot, MongoDB, REST API

Easy Park - Smart Car Parking System (IOT Project)

UOK

- Designed and built a prototype with sensors to detect available parking slots and update their status in real-time.
- Developed a mobile application that displays slot availability to users, enabling them to easily locate and reserve parking spaces.
- **Tech Stack:** Arduino, C, IoT Sensors, Embedded Systems

BIN MATRIX - Smart Waste Management System (Loading..)

- Developing a smart dustbin that can auto-segregate waste, a mobile app, and a web app for municipal councils connecting the smart bins and the general public.
- **Tech Stack:** ESP32, ESP32-CAM, MQ Gas Sensors, Moisture Sensors, Servo/Stepper Motors, OLED/LED Matrix Display, C++ (Arduino Framework), TensorFlow Lite for Microcontrollers, Hybrid Rule-Based Logic, MQTT/HTTP, Flutter, React.

SKILLS

- **Programming Languages:** Java, C, C#, JavaScript, SQL
- **Web Development:** HTML, CSS, Tailwind CSS, ReactJS, Spring Boot
- **Backend Services:** RESTful APIs, Spring Framework, MQTT, WebSockets, Postman, Swagger
- **Databases:** MySQL, MongoDB
- **IoT & Hardware:** ESP32, Arduino, Firmware Development, MQTTX, Concurrency Testing
- **AI & Productivity:** AI Architectures (RAG, Planning, Agentic Workflows), GitHub Copilot, Claude
- **Cloud:** GCP
- **Deployment:** CI/CD Pipelines, GitHub Actions, Git, GitHub
- **Tools & Platforms:** IntelliJ IDEA, VS Code, Antigravity, Figma, Grafana, Docker

PUBLICATIONS

Federated Learning for Privacy-Preserving Cyber Bullying Detection - Abstract Publication

5th Global Conference on Children and Youth, Oxford, UK – Mar 2025

[Link: [Abstract](#)]

Co-authors: D.S.P. Shashitha Udes De Silva, R. Sithumi Bhagya

- Proposed federated learning techniques to preserve children’s privacy in AI-powered cyberbullying detection systems.

Privacy Preserving Cyber Bullying Detection using Federated Learning - Abstract Publication

International Conference on Child Protection (ICCP) 2025, University of Kelaniya, Sri Lanka - Aug 2025

Co-authors: D.S.P. Shashitha Udes De Silva, R. Sithumi Bhagya

- Integrated privacy-preserving techniques to mitigate data leakage risks, ensuring AI-powered detection systems comply with sensitive data protection standards for children.

AWARDS AND COMMUNITY OUTREACH

Winner	SDG SPRINTS CHALLENGE	IEEE Sri Lanka Section SIGHT	2026
1st Runners up	HackX 10.0	Department of Industrial Management, UOK	2025
Finalists	CodeRally 4.0	IEEE Student Branch, IIT	2023
Finalists	League of Leaders'23	Commerce Club, UOK	2023

EXTRA CURRICULAR ACTIVITIES

President Humane Society of University of Kelaniya	2024 - 2025
Vice President Humane Society of University of Kelaniya	2023 - 2024
Secretary KelaniSTEAM of University of Kelaniya	2023 - 2024
Industry Relations Lead Electronics and Computer Science Club of University of Kelaniya	2023 - 2024
Organizing Committee Vice President for Partnership Development - CEO 2.0 AIESEC in University of Kelaniya	2023
Moderator at University Events UOK Robot Battles 2k24, AGMs, Night Sky Observation Camp (Astronomy and Space Science Association), Techsymphony : Online Tech workshops (ECS Club)	2022 - 2025

REFERENCES

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